SPARKBILLS · SPARKCURV TECHNOLOGIES

Sparkbills

Production Deployment Guide — AWS EC2 Ubuntu 24 + Nginx

Step-by-step, copy-pastable production install of Sparkbills on a fresh AWS EC2 instance running Ubuntu 24.04 LTS. Covers instance sizing, security group rules, MongoDB 7 install, backend systemd service, React build, Nginx reverse-proxy with HTTPS, Let's Encrypt, backups to S3 and the recommended upgrade procedure. Every command is followed by the expected output so you can catch mistakes early.

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Vendor	**SparkCurv Technologies Pvt. Ltd.**
Contact	**admin@sparkcurv.in**
Audience	**Business owners, IT admins, DevOps**

Contents

1. Prerequisites

- An AWS account with IAM permission to create EC2, Security Groups and Elastic IP.
- A registered domain name, e.g. *billing.sparkcurv.in*, with editable DNS.
- An SSH key-pair (.pem file) already downloaded to your laptop.
- The Sparkbills source code either as a Git repo or a ZIP archive.
- A machine to SSH from — Windows (PuTTY / OpenSSH), macOS, or Linux.

2. Choose the right EC2 size (capacity planning)

Users online	Instance	vCPU / RAM	Storage	Est. cost / month
1 – 3	t3.small	2 / 2 GB	20 GB gp3	≈ $18
3 – 10	t3.medium	2 / 4 GB	30 GB gp3	≈ $34
10 – 30	t3.large	2 / 8 GB	50 GB gp3	≈ $66
30 – 100	m6i.large + separate DB	2 / 8 GB + 4 / 16 GB	100 GB	≈ $180

Note: t3.medium is the sweet spot for most SMBs. Start there and scale up later.

3. Launch and access the EC2 instance

From the AWS console: EC2 → Launch Instance:

- **AMI**: Ubuntu Server 24.04 LTS (HVM), SSD, 64-bit x86.
- **Instance type**: as per the capacity table above.
- **Key pair**: your existing .pem or create a new one.
- **Storage**: root volume 30 GB gp3 (3000 IOPS, 125 MB/s).
- **Security Group**: inbound 22 (SSH from your IP), 80 (HTTP anywhere), 443 (HTTPS anywhere). Do NOT open 27017 — MongoDB stays on 127.0.0.1.
- Attach a fresh Elastic IP so the public IP survives reboots.
- In your DNS panel, add an A record: billing.sparkcurv.in → the Elastic IP. TTL 300 s.

From your laptop:

```
chmod 400 sparkbills-key.pem
ssh -i sparkbills-key.pem ubuntu@&lt;ELASTIC-IP&gt;

# Expected: welcome banner and prompt like
# ubuntu@ip-10-0-1-42:~$
```

4. Update Ubuntu and set the hostname / timezone

```
sudo apt update && sudo apt upgrade -y
sudo timedatectl set-timezone Asia/Kolkata
sudo hostnamectl set-hostname sparkbills-prod
exec bash   # reload prompt

# Verify
timedatectl | grep 'Time zone'
# Time zone: Asia/Kolkata (IST, +0530)
```

5. Install Python 3, Node 20, Nginx and Certbot

```
# Python 3.12 + build tools + git
sudo apt install -y python3 python3-venv python3-pip build-essential git curl wget

# Node.js 20 LTS via NodeSource
curl -fsSL https://deb.nodesource.com/setup_20.x | sudo -E bash -
sudo apt install -y nodejs
sudo npm install -g yarn

# Verify versions (expected output shown after &lt; &gt;)
python3 --version    # &lt; Python 3.12.x &gt;
node -v              # &lt; v20.x.x &gt;
yarn -v              # &lt; 1.22.x &gt;

# Nginx + Certbot
sudo apt install -y nginx
sudo snap install --classic certbot
sudo ln -s /snap/bin/certbot /usr/bin/certbot

# Firewall &mdash; only HTTP/HTTPS + SSH
sudo ufw allow OpenSSH
sudo ufw allow 'Nginx Full'
sudo ufw --force enable
sudo ufw status
# Status: active
```

6. Install MongoDB 7 Community

```
sudo apt install -y gnupg
curl -fsSL https://pgp.mongodb.com/server-7.0.asc | \
  sudo gpg -o /usr/share/keyrings/mongodb-server-7.0.gpg --dearmor

echo "deb [ arch=amd64,arm64 signed-by=/usr/share/keyrings/mongodb-server-7.0.gpg ] \
  https://repo.mongodb.org/apt/ubuntu noble/mongodb-org/7.0 multiverse" | \
  sudo tee /etc/apt/sources.list.d/mongodb-org-7.0.list

sudo apt update
sudo apt install -y mongodb-org
sudo systemctl enable --now mongod

# Verify (should show 'active (running)')
sudo systemctl status mongod --no-pager | head -3
mongosh --eval 'db.version()'
# 7.0.x
```

Important: MongoDB binds to 127.0.0.1:27017 by default. Do **NOT** change this and do **NOT** open port 27017 in your Security Group. The FastAPI backend runs on the same instance and connects locally.

7. Clone Sparkbills source into /opt/sparkbills

```
sudo mkdir -p /opt/sparkbills
sudo chown ubuntu:ubuntu /opt/sparkbills
cd /opt/sparkbills
git clone https://&lt;YOUR-REPO-URL&gt;.git .
# Or, if the vendor gave you a ZIP:
#   scp -i ~/sparkbills-key.pem sparkbills.zip ubuntu@&lt;ELASTIC-IP&gt;:~/
#   unzip ~/sparkbills.zip -d /opt/sparkbills
ls
# backend  frontend  deploy  scripts  ...
```

8. Backend setup — venv, dependencies, .env

```
cd /opt/sparkbills/backend
python3 -m venv venv
source venv/bin/activate
pip install --upgrade pip
pip install -r requirements.txt
pip install emergentintegrations --extra-index-url \
  https://d33sy5i8bnduwe.cloudfront.net/simple/
```

Now create /opt/sparkbills/backend/.env with the following contents. Replace values in ALL-CAPS < ... > with your own:

```
MONGO_URL="mongodb://localhost:27017"
DB_NAME="sparkbills_prod"
CORS_ORIGINS="https://billing.sparkcurv.in"

# Generate a fresh secret using the command below the block
JWT_SECRET="&lt;64-char-hex-generated-in-next-step&gt;"

# Seeded owner on first start
ADMIN_EMAIL="admin@sparkcurv.in"
ADMIN_PASSWORD="Admin@2026"
ADMIN_NAME="SparkCurv Admin"
ADMIN_BUSINESS_NAME="SparkCurv Technologies"

# Email (invoice PDF to customer). Leave empty to disable.
EMERGENT_EMAIL_KEY="&lt;your-emergent-email-key&gt;"
EMAIL_FROM_NAME="Sparkbills"
```

Generate the JWT secret (do this on the server):

```
python3 -c "import secrets;print(secrets.token_hex(32))"
# 64-character hex string &mdash; paste into JWT_SECRET
```

Note: The .env file must have chmod 600. It contains the DB URI and JWT secret — treat it like a password file.

9. Systemd service for the backend

Create /etc/systemd/system/sparkbills-backend.service:

```
[Unit]
Description=Sparkbills FastAPI backend
After=network.target mongod.service
Wants=mongod.service

[Service]
Type=simple
```

```
User=ubuntu
Group=ubuntu
WorkingDirectory=/opt/sparkbills/backend
EnvironmentFile=/opt/sparkbills/backend/.env
ExecStart=/opt/sparkbills/backend/venv/bin/uvicorn server:app \
  --host 127.0.0.1 --port 8001 --workers 2
Restart=always
RestartSec=5
StandardOutput=append:/var/log/sparkbills-backend.log
StandardError=append:/var/log/sparkbills-backend.err

[Install]
WantedBy=multi-user.target
```

```
sudo touch /var/log/sparkbills-backend.log /var/log/sparkbills-backend.err
sudo chown ubuntu:ubuntu /var/log/sparkbills-backend.*

sudo systemctl daemon-reload
sudo systemctl enable --now sparkbills-backend

# Verify
sudo systemctl status sparkbills-backend --no-pager | head -5
# active (running)

curl http://127.0.0.1:8001/api/
# {"app":"Sparkbills","vendor":"SparkCurv Technologies","status":"ok"}
```

10. Frontend build (React → static bundle)

Create /opt/sparkbills/frontend/.env:

```
REACT_APP_BACKEND_URL="https://billing.sparkcurv.in"
```

Then build:

```
cd /opt/sparkbills/frontend
yarn install --frozen-lockfile
yarn build
# Compiled successfully.
# The build folder is ready to be deployed.

ls build/ | head -5
# asset-manifest.json  brand  docs  index.html  static/
```

11. Nginx reverse proxy (HTTP + HTTPS)

Create /etc/nginx/sites-available/sparkbills:

```
# HTTP -> HTTPS redirect
server {
    listen 80;
    server_name billing.sparkcurv.in;
    return 301 https://$host$request_uri;
}

# HTTPS main
server {
    listen 443 ssl http2;
    server_name billing.sparkcurv.in;

    # Certificates filled in by certbot in step 12
    ssl_certificate     /etc/letsencrypt/live/billing.sparkcurv.in/fullchain.pem;
    ssl_certificate_key /etc/letsencrypt/live/billing.sparkcurv.in/privkey.pem;
    ssl_protocols TLSv1.2 TLSv1.3;
    ssl_ciphers HIGH:!aNULL:!MD5;

    client_max_body_size 25M;   # for invoice attachments

    # React SPA static bundle
    root /opt/sparkbills/frontend/build;
    index index.html;
    location / {
        try_files $uri /index.html;   # SPA fallback
    }

    # FastAPI backend
    location /api/ {
        proxy_pass         http://127.0.0.1:8001/api/;
        proxy_http_version 1.1;
        proxy_set_header   Host $host;
        proxy_set_header   X-Real-IP $remote_addr;
        proxy_set_header   X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header   X-Forwarded-Proto $scheme;
        proxy_read_timeout 60s;
    }

    gzip on;
    gzip_types text/plain text/css application/json application/javascript image/svg+xml;
    gzip_min_length 512;
```

```
}

sudo ln -s /etc/nginx/sites-available/sparkbills /etc/nginx/sites-enabled/
sudo rm -f /etc/nginx/sites-enabled/default

# test config
sudo nginx -t
# nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
# nginx: configuration file /etc/nginx/nginx.conf test is successful

sudo systemctl reload nginx
```

12. Enable HTTPS via Let's Encrypt

```
sudo certbot --nginx -d billing.sparkcurv.in \
  --agree-tos --non-interactive --email admin@sparkcurv.in --redirect

# Successfully deployed certificate for billing.sparkcurv.in
```

Certbot installs a systemd timer to renew every 60 days. Verify:

```
sudo systemctl list-timers | grep certbot
# certbot.timer  ...  snap.certbot.renew.service
```

13. First-login smoke test

1. Visit **https://billing.sparkcurv.in**. The animated login page loads.
2. Sign in with the seeded owner: admin@sparkcurv.in / Admin@2026.
3. You land on the Dashboard with all KPIs at 0.
4. Open **Settings** → **Business**, fill in your GSTIN and UPI ID.
5. Open **Settings** → **Users**, click your row → *Change Password*. Set a passphrase.
6. Create a test invoice from **Sales** → **New Invoice**. Confirm PDF opens.

14. Daily backups to S3

```
sudo apt install -y awscli
aws configure   # supply access key of a restricted IAM user

# /opt/sparkbills/backup.sh
cat &gt; /opt/sparkbills/backup.sh &lt;&lt;'EOS'
#!/bin/bash
STAMP=$(date +%Y%m%d-%H%M)
OUT=/tmp/sb-$STAMP.gz
mongodump --db sparkbills_prod --archive=$OUT --gzip
aws s3 cp $OUT s3://YOUR-BUCKET/sparkbills/
rm $OUT
EOS

chmod +x /opt/sparkbills/backup.sh

# Every day at 02:30 local time
(crontab -l 2&gt;/dev/null; echo '30 2 * * * /opt/sparkbills/backup.sh') | crontab -
```

Note: Set an S3 lifecycle rule to move backups older than 30 days to Glacier Deep Archive — keeps GST-audit-mandated 8-year retention at < $0.001/GB/mo.

15. Upgrade procedure (zero-downtime-ish)

```
cd /opt/sparkbills
git pull

# Backend
cd backend
source venv/bin/activate
pip install -r requirements.txt
sudo systemctl restart sparkbills-backend

# Frontend
cd ../frontend
yarn install --frozen-lockfile
yarn build
sudo systemctl reload nginx     # picks up the new static bundle
```

Note: Downtime is only during the < 3-second backend restart. Sessions (JWTs) survive.

16. Security hardening checklist

- Root SSH login disabled — only ubuntu user.
- Change SSH port from 22 → 2222 (optional) — update SG.

- Install fail2ban: sudo apt install -y fail2ban.
- Enforce unattended-upgrades: sudo dpkg-reconfigure --priority=low unattended-upgrades.
- Set MongoDB auth (if you ever expose it): edit /etc/mongod.conf and enable security.authorization: enabled.
- Rotate the JWT_SECRET every 6 months (users will need to re-login).
- Enable AWS CloudWatch agent for CPU / disk alerts.

17. Monitoring and log tailing

```
# Backend logs (live)
sudo journalctl -u sparkbills-backend -f

# Backend logs (last 200 lines)
sudo journalctl -u sparkbills-backend -n 200 --no-pager

# Nginx access + error
sudo tail -f /var/log/nginx/access.log
sudo tail -f /var/log/nginx/error.log

# Mongo
sudo tail -f /var/log/mongodb/mongod.log
```

18. Common issues and fixes

Symptom	Likely cause	Fix
Blank page + 502 from /api/	Backend service is down	sudo journalctl -u sparkbills-backend -n 100 — look for tracebacks; restart service.
Login POST returns HTML	CORS mismatch	Ensure CORS_ORIGINS in backend/.env exactly matches your https domain (including scheme).
MongoDB refused connection	mongod not running	sudo systemctl status mongod; sudo systemctl start mongod.
Certificate renewal fails	Port 80 blocked in SG	Open port 80 in the Security Group — Certbot uses HTTP-01 challenge.
Refresh of /app/... returns 404	Nginx SPA fallback missing	Ensure 'try_files $uri /index.html;' is present in the / block.
Emails stuck in 'failed'	EMERGENT_EMAIL_KEY invalid	Contact SparkCurv to reissue key; sparkbills gracefully falls back to HTML-only.

For 24x7 SLA support contact **admin@sparkcurv.in** with your EC2 instance ID, domain and a journalctl excerpt.